# Thanh Lam

Machine Learning Researcher

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I am a machine learning researcher interested in the theoretical foundations and practical development of learning algorithms. I work as a Quantitative Researcher at Citadel Securities, where I spearhead neural-network research for the high-frequency futures team, developing deep-learning methods for short-horizon time-series prediction. I received my PhD in Computer Science from the National University of Singapore, advised by Prof. Bryan Low (NUS) and Prof. Patrick Jaillet (MIT), working on reinforcement learning, sequential decision-making, and learning theory. My research has appeared at ICLR, ICML, and TMLR.

## RESEARCH EXPERIENCE

**Quantitative Researcher** · Citadel Securities, London — 2023 – Present

- Spearhead the high-frequency futures team's neural-network alpha research.
- Built the team's deep-learning training recipe from scratch to production: large-scale sequential recurrent models for short-horizon prediction on noisy, non-stationary market data.

**Quantitative Research Intern** · Citadel Securities, Singapore — 2022

Alpha research in digital assets.

## PUBLICATIONS

### Peer-reviewed

[1] **Risk-Aware Reinforcement Learning with Coherent Risk Measures and Non-linear Function Approximation. Thanh Lam**, Arun Verma, Bryan Low, Patrick Jaillet. *ICLR 2023.*

[2] **On Average-Case Error Bounds for Kernel-Based Bayesian Quadrature.** Xu Cai*, **Thanh Lam***, Jonathan Scarlett. *TMLR 2023.*

[3] **Model Fusion for Personalized Learning. Thanh Lam***, Trong Nghia Hoang*, Bryan K. H. Low, Patrick Jaillet. *ICML 2021.*

[4] **Learning Task-Agnostic Embedding of Multiple Black-Box Experts for Multi-Task Model Fusion.** Trong Nghia Hoang*, **Thanh Lam***, Bryan K. H. Low, Patrick Jaillet. *ICML 2020.*

### Preprints & working papers

[5] **Implicit Bias via Spectral Neural Networks and Non-linear Matrix Sensing.** Hong T. M. Chu, Xiaoyu Dong, Subhro Ghosh, **Thanh Lam***, Soumendu Sundar Mukherjee. *arXiv:2402.17595, 2024.*

[6] **Online Data Procurement with Self-Interested Agents. Thanh Lam**, Bryan Low, Patrick Jaillet. *Working paper.*

## EDUCATION

**Ph.D., Computer Science** · National University of Singapore — 2019 – 2023

Advised by Prof. Bryan Low (NUS) and Prof. Patrick Jaillet (MIT). GPA 4.6/5.

**Visiting Graduate Student** · MIT, Laboratory for Information & Decision Systems (LIDS) — Fall 2022

Hosted by Prof. Patrick Jaillet.

**B.Comp., Computer Science (First Class Honours)** · National University of Singapore — 2015 – 2019

GPA 4.6/5.

## RESEARCH AREAS

Theoretical machine learning, reinforcement learning, sequential decision-making, multi-armed bandits, optimization, deep learning, statistical learning, Bayesian methods.

## AWARDS & SERVICE

**Awards:** SMART Graduate Fellowship (sole recipient across 4 universities, 2019); Research Achievement Award (×2, 2020–21); Honor List for Teaching Excellence (2021); Dean's List (2019); ASEAN Undergraduate Scholarship (1 of 3 nationwide, 2015).

**Service:** Reviewer for ICML, NeurIPS, ICLR, AISTATS (2021–2023), Neural Networks, and IEEE RA-L; volunteer at ICLR, ICML, and NeurIPS (2020–2021).